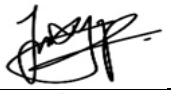
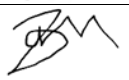
UNIVERSITI MALAYSIA SARAWAK (UNIMAS)
Faculty of Computer Science and Information

Premier Digital Tech Institution

Assignment/Report Cover Sheet

Student Name	Student ID Number	Group No	Signature
CHAN KAR KAY	103618	G3	
JENX YAP YEE XUAN	104527	G3	
NURZAMARUL ISLAM BIN NURDDIN	105977	G3	

Subject Code :TMF2034	**Subject Name :Database Concept and Design**
Assignment Title :FitLife Wellness Centre Database System	**Lecturer :Dr Bong Chih How**
Due Date :5/1/2026	**Date Submitted :6/1/206**

This cover sheet must be completed, signed and firmly attached to the front of the submission. All work must be submitted by the due date. If an extension of work is granted, an assignment extension acknowledgement slip must be signed by the lecturer/tutor and attached to assignment. Please note that is your responsibility to retain copies of your assignment.

Plagiarism and Collusion are methods of cheating that falls under Peraturan Akademik Universiti Malaysia Sarawak para 11: Etika Akademik

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Collusion
Collusion is the presentation of work that is the result in whole or in part of unauthorized collaboration with another person or persons.

Where there are reasonable grounds for believing that cheating has occurred, the only action that may be taken when plagiarism or collusion is detected is for the staff member not to mark the item of work and to report or refer the matter to the Dean. This may result in work being disallowed and given a fail grade or if the circumstances warrant, the matter may be referred to a Committee of inquiry for investigation. Such investigation may result in the matter being referred to the University Discipline Committee, **which** has the power to exclude a student.

MARK :

1. Relational Database Diagram

1.1. Conversion of ERD into Relational Database Schema

Core Entities Conversion
Trainer

Attribute	Key
Trainer_ID	PK
Trainer_Name	
Trainer_Specialization	
Trainer_Certification	
Trainer_Contact	

Program

Attribute	Key
Program_ID	PK
Program_Name	
Program_Duration	
Program_Category	

Class

Attribute	Key
Class_ID	PK
Program_ID	FK
Room_ID	FK
Trainer_ID	FK
Class_Name	
Class_Date	
Start_Time	
End_Time	
Max_Capacity	
Class_Type	
Class_Status	

Other Entities Conversion
Member

Attribute	Key
Member_ID	PK
Member_Name	
Member_Contact	
Member_DOB	
Member_Gender	
Membership_Type_ID	FK

Membership_Status_ID	FK

Membership_Types

Attribute	Key
Membership_Type_ID	PK
Membership_Type_Name	
Membership_Description	
Membership_Fee	

Membership_Status

Attribute	Key
Membership_Status_ID	PK
Membership_Status	
Status_Description	

Enrollment

Attribute	Key
Enrollment_ID	PK
Member_ID	FK
Program_ID	FK
Enrollment_Date	

Payment

Attribute	Key
Payment_ID	PK
Member_ID	FK
Membership_Type_ID	FK
Payment_Date	
Payment_Amount	
Payment_Method	

Room

Attribute	Key
Room_ID	PK
Room_Name	
Room_Capacity	

1.2. Time-Variant Data Management

Salary_History

Attribute	Key
Salary_ID	PK
Trainer_ID	FK
Salary_Amount	
Salary_Date	
Salary_Note	

Trainer_Program_History (New Entity)

Attribute	Key
History_ID	PK
Trainer_ID	FK
Program_ID	FK
Assignment_Date	
Change_Reason	

This table enables the system to store historical records without overwriting existing data.

1.3. Normalization of the Database

1.3.1. First Normal Form (1NF)

Requires that all attributes contain atomic values and that no table has repeating groups.

- The table in the schema complies with 1NF.
- Example, the *Member* table does not store multiple program enrollments in a single field; instead, the *Enrollment* associative table manages this relationship.
- All tables are stored as single value, indivisible attributes with no repeating groups.

1.3.2. Second Normal Form (2NF)

Requires that all non-key attributes are fully functionally dependent on the entire primary key, eliminating partial dependencies.

- All current tables use single-attribute primary keys, there are no partial dependencies
- Associative table *Enrollment* attributes like *Enrollment_Date* depend entirely on the primary key *Enrollment_ID.*
- All tables satisfy 2NF, no attribute depends only on part of a composite key.

1.3.3. Third Normal Form (3NF)

Eliminates transitive dependencies, where a non-key attribute depends on another non-key attribute rather than the primary key.

- Attributes that describe categories or types are moved to lookup tables (small, separate table used to store reference data)
    - *Membership_Types* and *Membership_Status* store descriptive membership information referenced by *Member*.
    - *Trainer_Specialization* and *Trainer_Certification* can be moved to separate tables

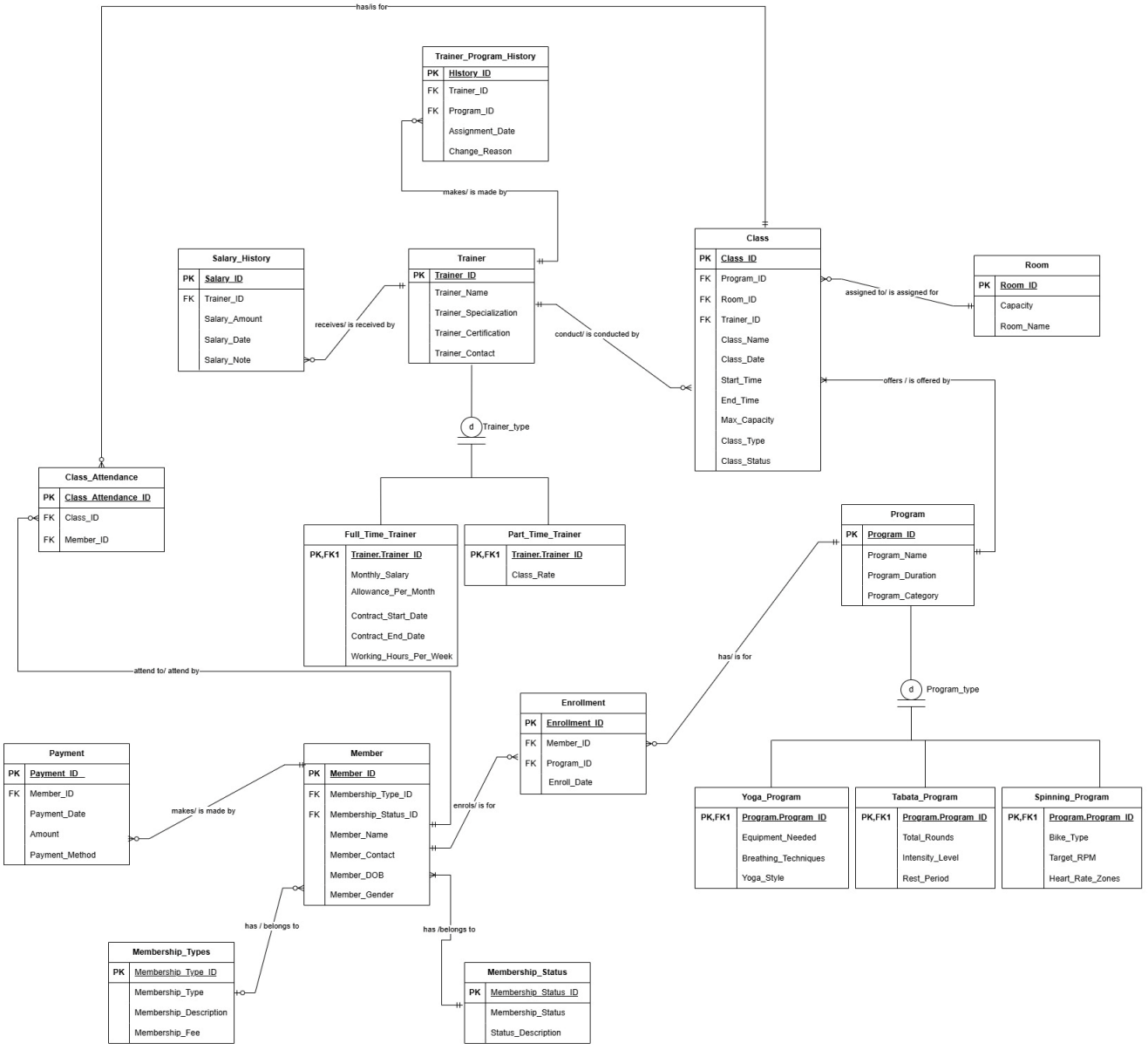
to prevent repetition across trainers.

- Time-variant data is stored in separate tables (Example: *Salary_History*) ensuring historical salary changes do not overwrite existing records.
- All non-key attributes depend solely on their respective primary keys, thus transitive dependencies are eliminated.

Anomoly Elimination

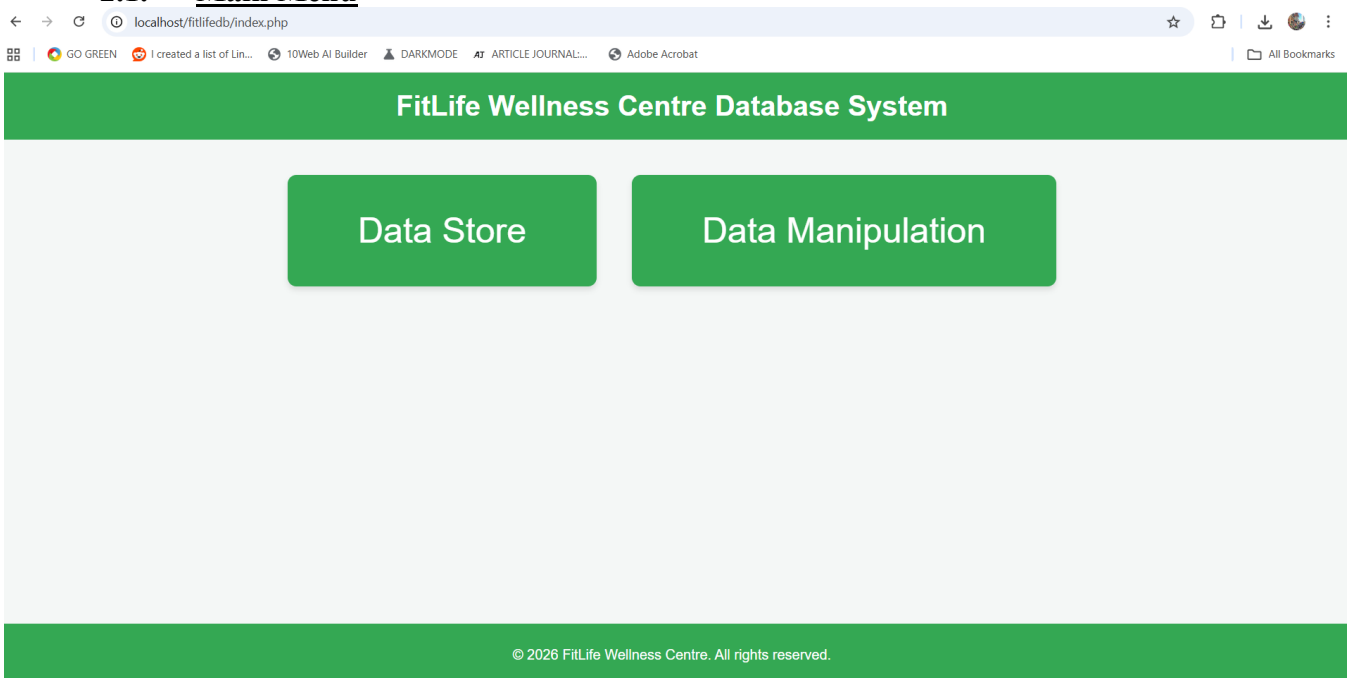
Anomaly type	Elimination
Insertion	Normalized tables enable the insertion of new programs or members without requiring dummy data.
Update	Only one location (lookup or history table) needs to be updated to be able to update membership types, trainer certifications or salary.
Deletion	Programs or historical data that are related are not automatically deleted when a member or class is removed.

1.4. Crow's Foot Diagram

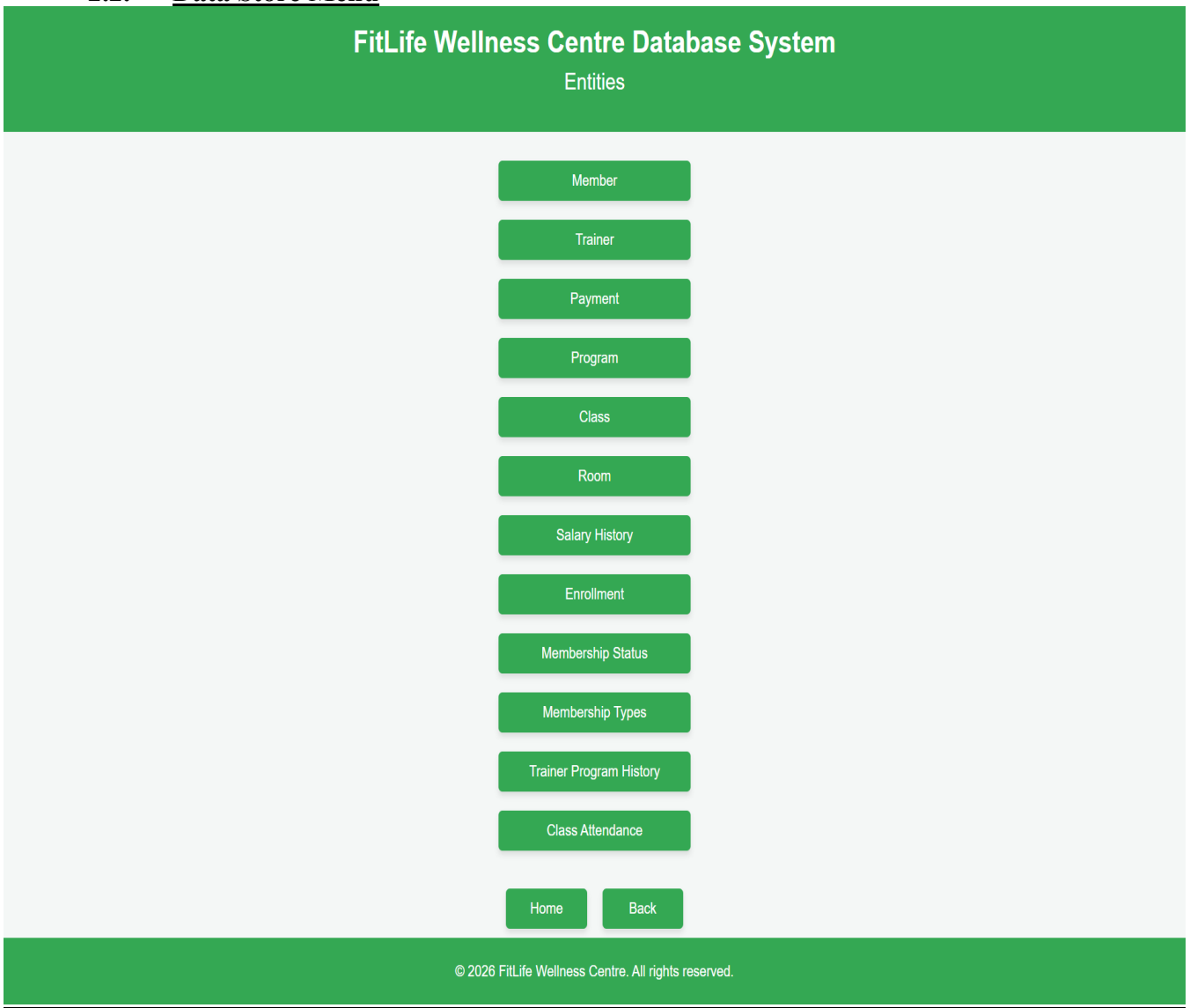


2. Imporatant Functions

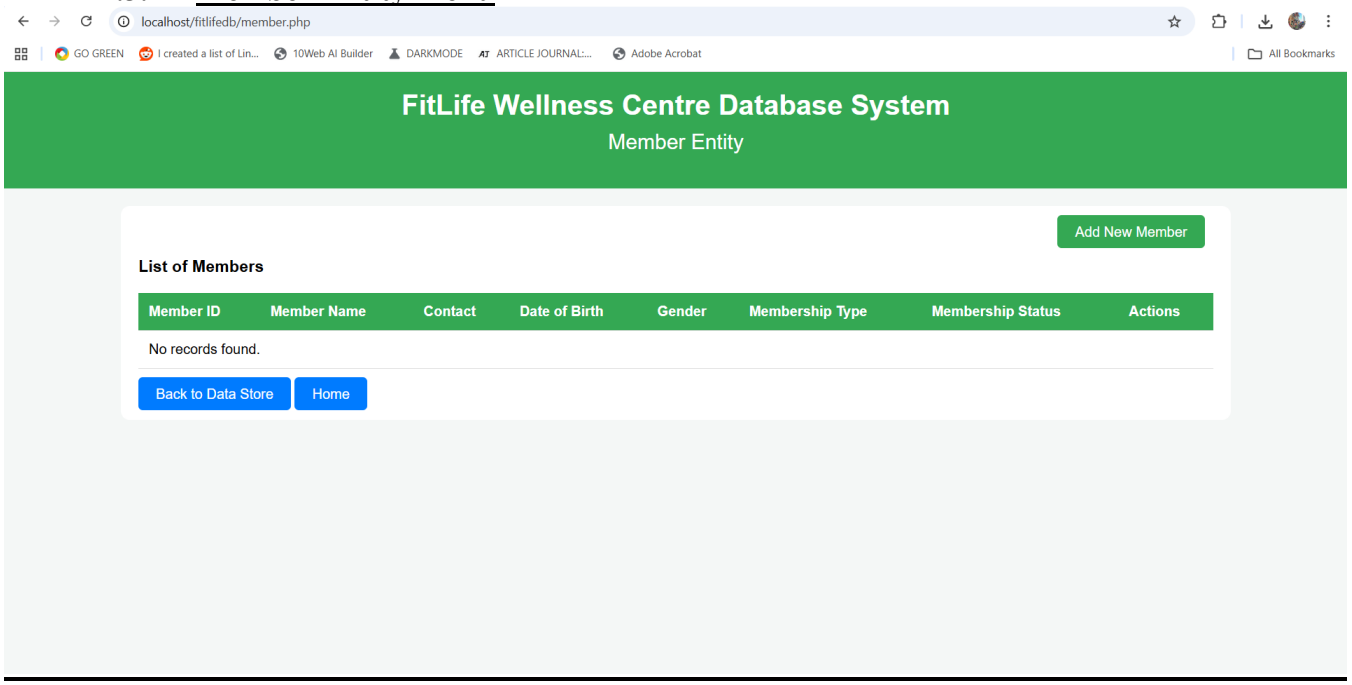
2.1. Main Menu



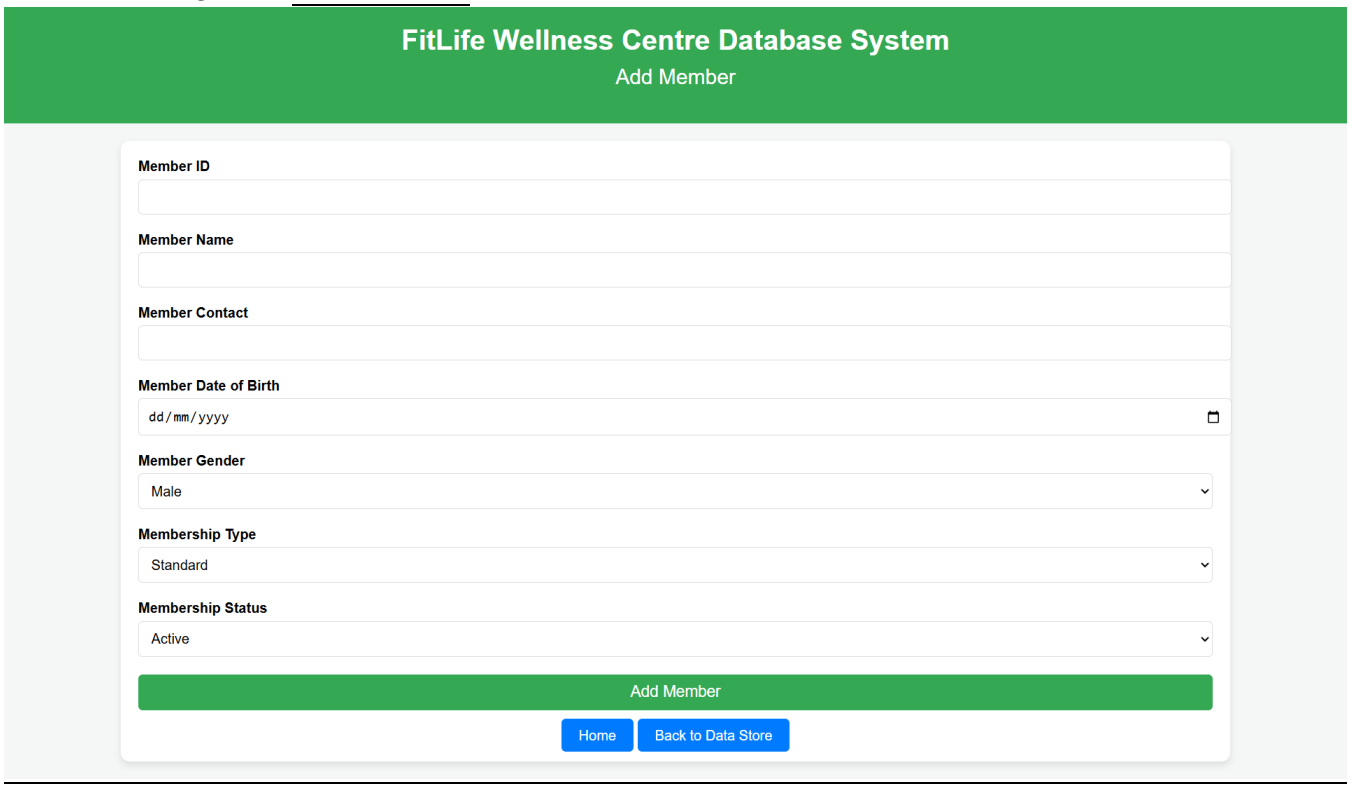
2.2. Data Store Menu



2.3. Member Entity Menu



2.3.1. Add Function



FitLife Wellness Centre Database System

Add Member

Member ID

1

Member Name

Divesh

Member Contact

0173132214

Member Date of Birth

12/12/2000

Member Gender

Male

Membership Type

Standard

Membership Status

Active

Add Member

Home | Back to Data Store

Membership Type

Standard

- Standard
- Premium
- Family
- Student
- Corporate

Home | Back to Data Store

Membership Status

Active

- Active
- Inactive
- Suspended
- Expired
- Pending

New member added successfully!

FitLife Wellness Centre Database System

Add Member

Member ID

FitLife Wellness Centre Database System

Member Entity

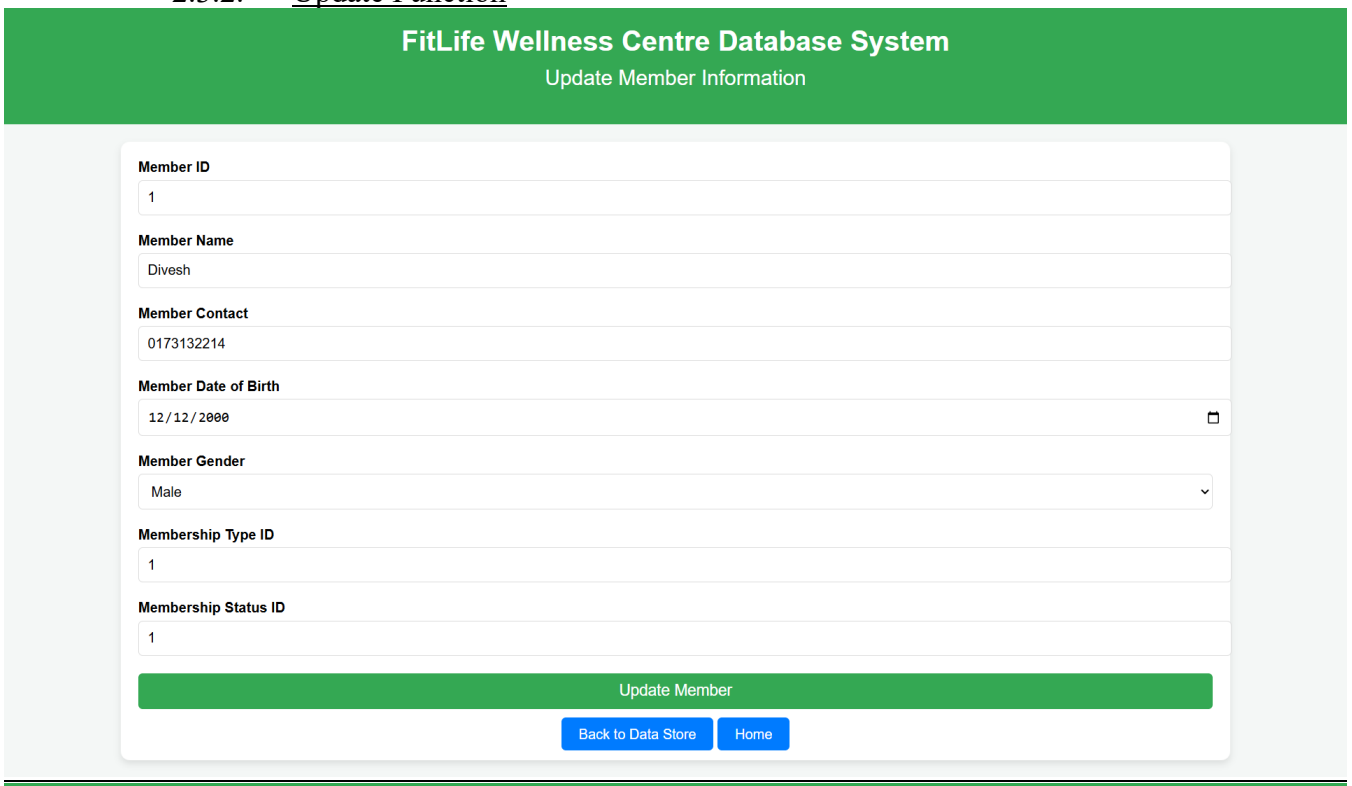
Add New Member

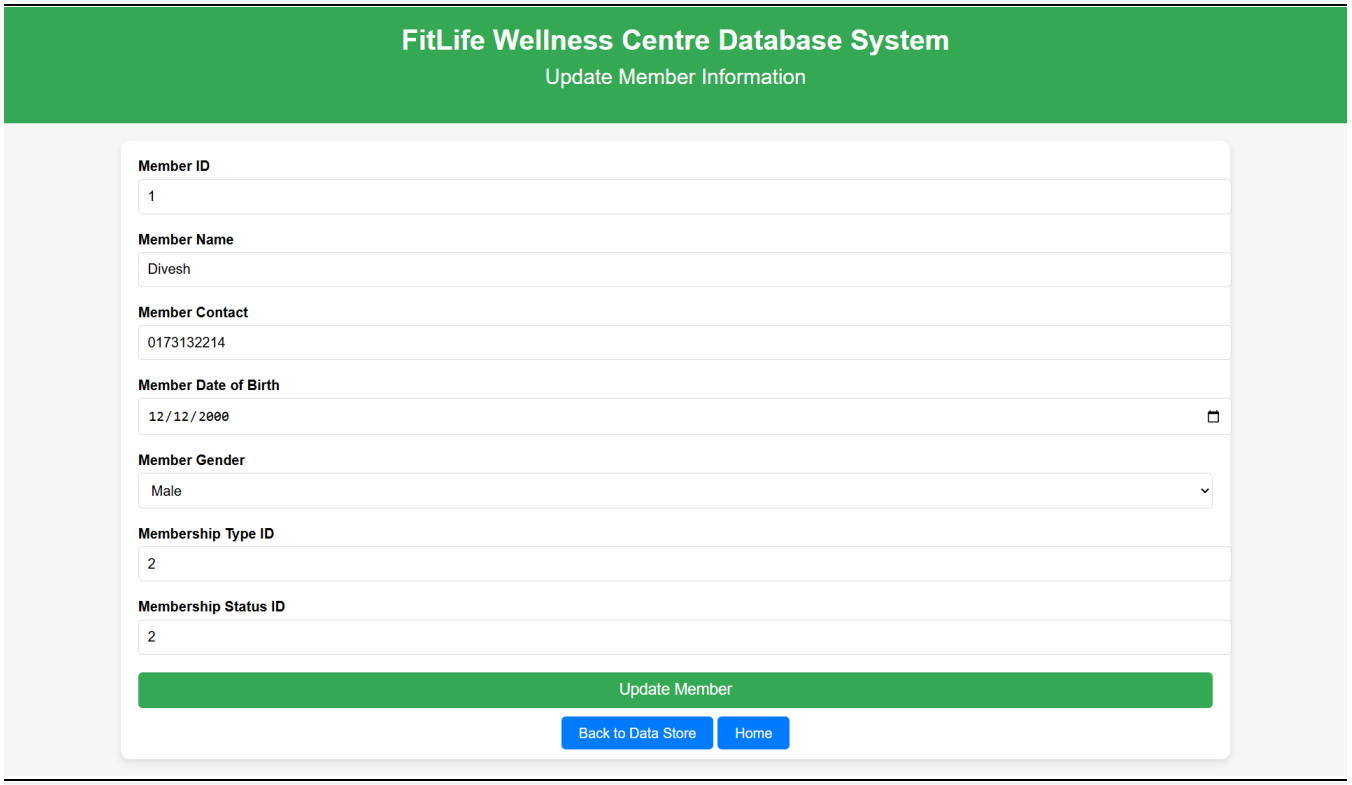
List of Members

Member ID	Member Name	Contact	Date of Birth	Gender	Membership Type	Membership Status	Actions
1	Divesh	0173132214	2000-12-12	Male	Standard	Active	Delete Update

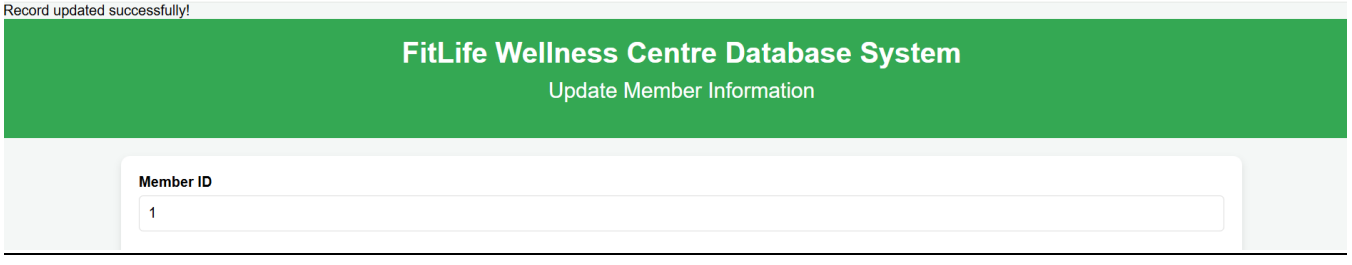
Back to Data Store | Home

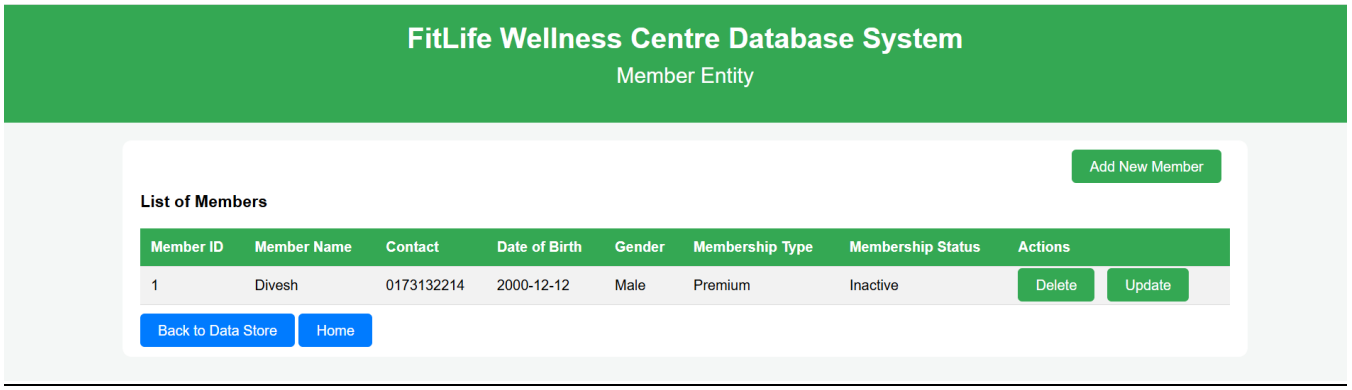
2.3.2. Update Function



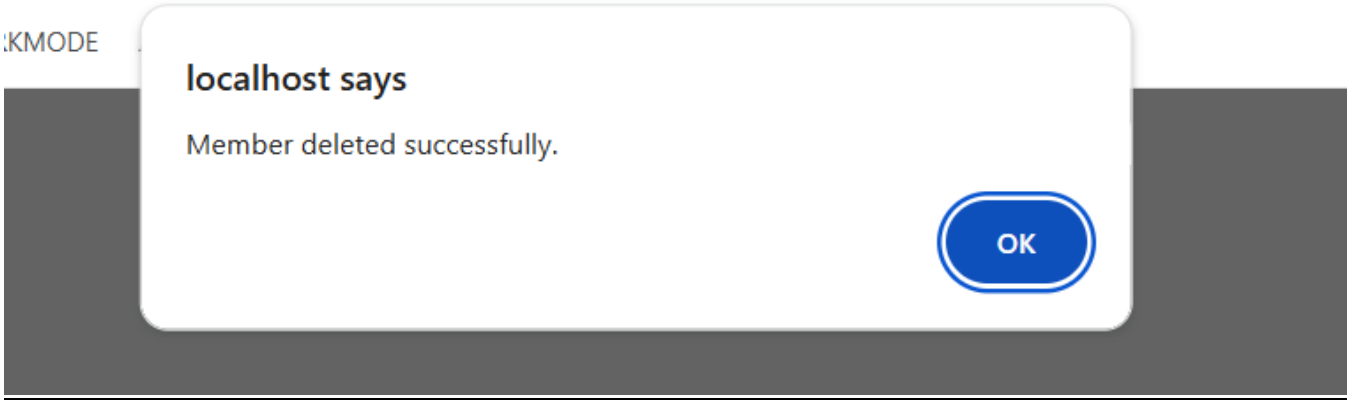


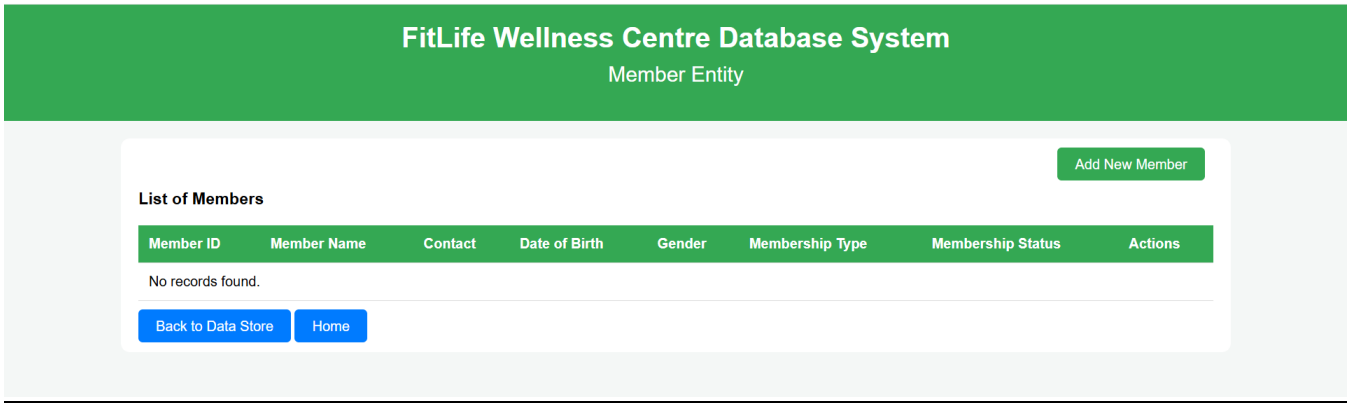
Record updated successfully!



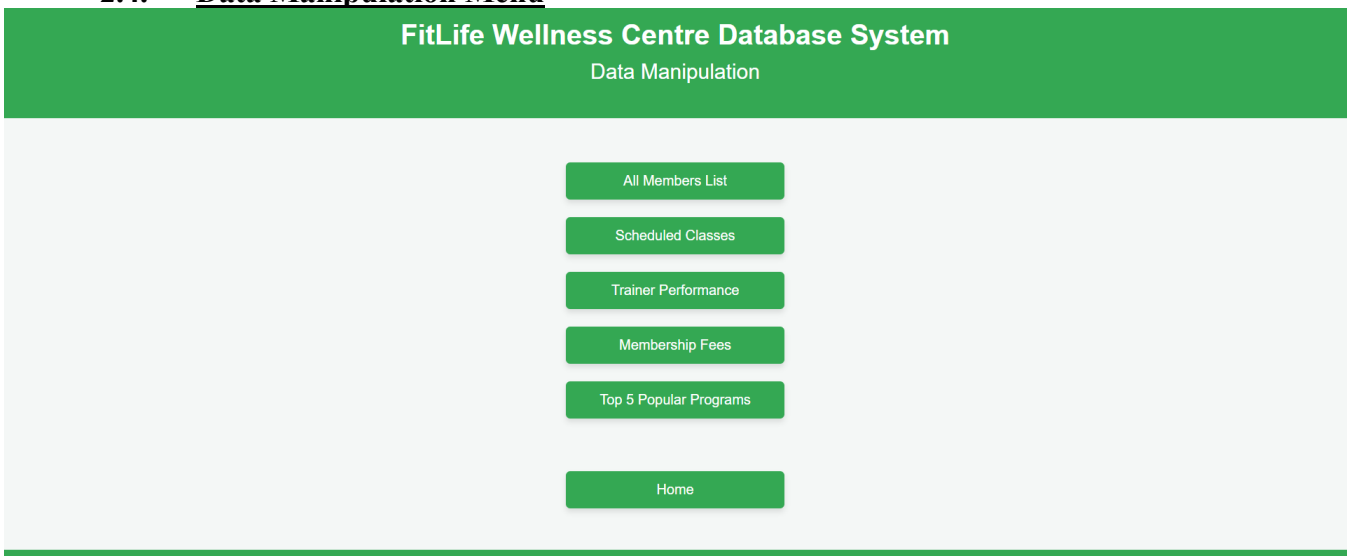


2.3.3. Delete Function

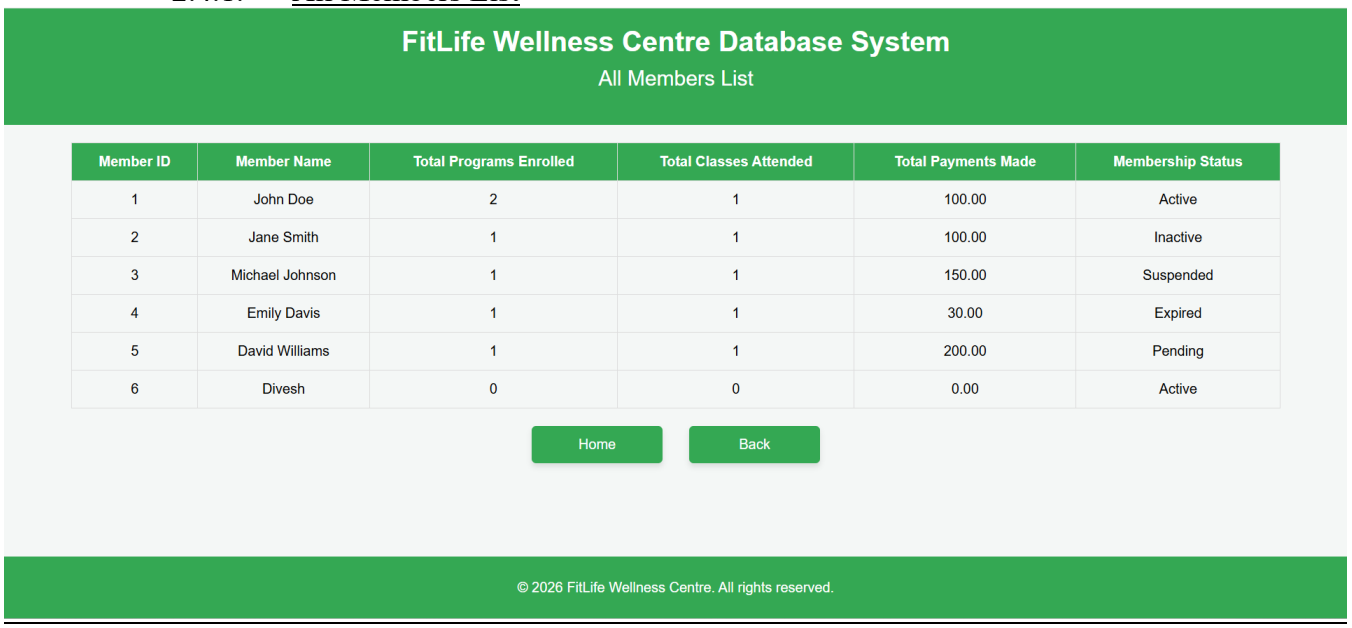




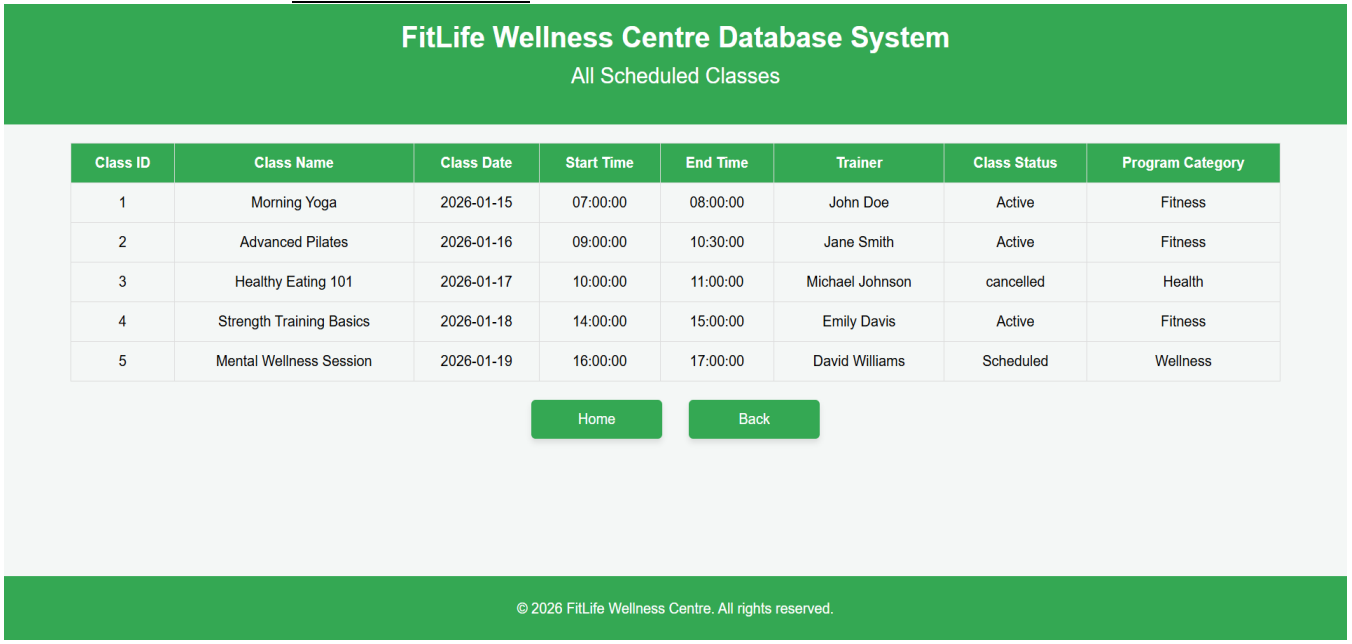
2.4. Data Manipulation Menu



2.4.1. All Members List



2.4.2. Scheduled Classes



2.4.3. Trainer Performance

FitLife Wellness Centre Database System

All Scheduled Classes

Class ID	Class Name	Class Date	Start Time	End Time	Trainer	Class Status	Program Category
1	Morning Yoga	2026-01-15	07:00:00	08:00:00	John Doe	Active	Fitness
2	Advanced Pilates	2026-01-16	09:00:00	10:30:00	Jane Smith	Active	Fitness
3	Healthy Eating 101	2026-01-17	10:00:00	11:00:00	Michael Johnson	cancelled	Health
4	Strength Training Basics	2026-01-18	14:00:00	15:00:00	Emily Davis	Active	Fitness
5	Mental Wellness Session	2026-01-19	16:00:00	17:00:00	David Williams	Scheduled	Wellness

Home Back

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2.4.4. Membership Fees

FitLife Wellness Centre Database System

Quarterly and Annual Membership Fees

Member Name	Membership Type	Monthly Fee (RM)	Quarterly Fee (RM)	Annual Fee (RM)
John Doe	Standard	50.00	150.00	600.00
Jane Smith	Premium	100.00	300.00	1,200.00
Michael Johnson	Family	150.00	450.00	1,800.00
Emily Davis	Student	30.00	90.00	360.00
David Williams	Corporate	200.00	600.00	2,400.00
Divesh	Standard	50.00	150.00	600.00

Home Back

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2.4.5. Top 5 Popular Programs

FitLife Wellness Centre Database System

Top 5 Most Popular Programs

Program Name	Total Enrolled Members	Assigned Trainer	Program Category
Advanced Pilates	2	Jane Smith	Fitness
Yoga Basics	1	John Doe	Fitness
Intro to Nutrition	1	Michael Johnson	Health
Strength Training	1	Emily Davis	Fitness
Mental Wellness Workshop	1	David Williams	Wellness

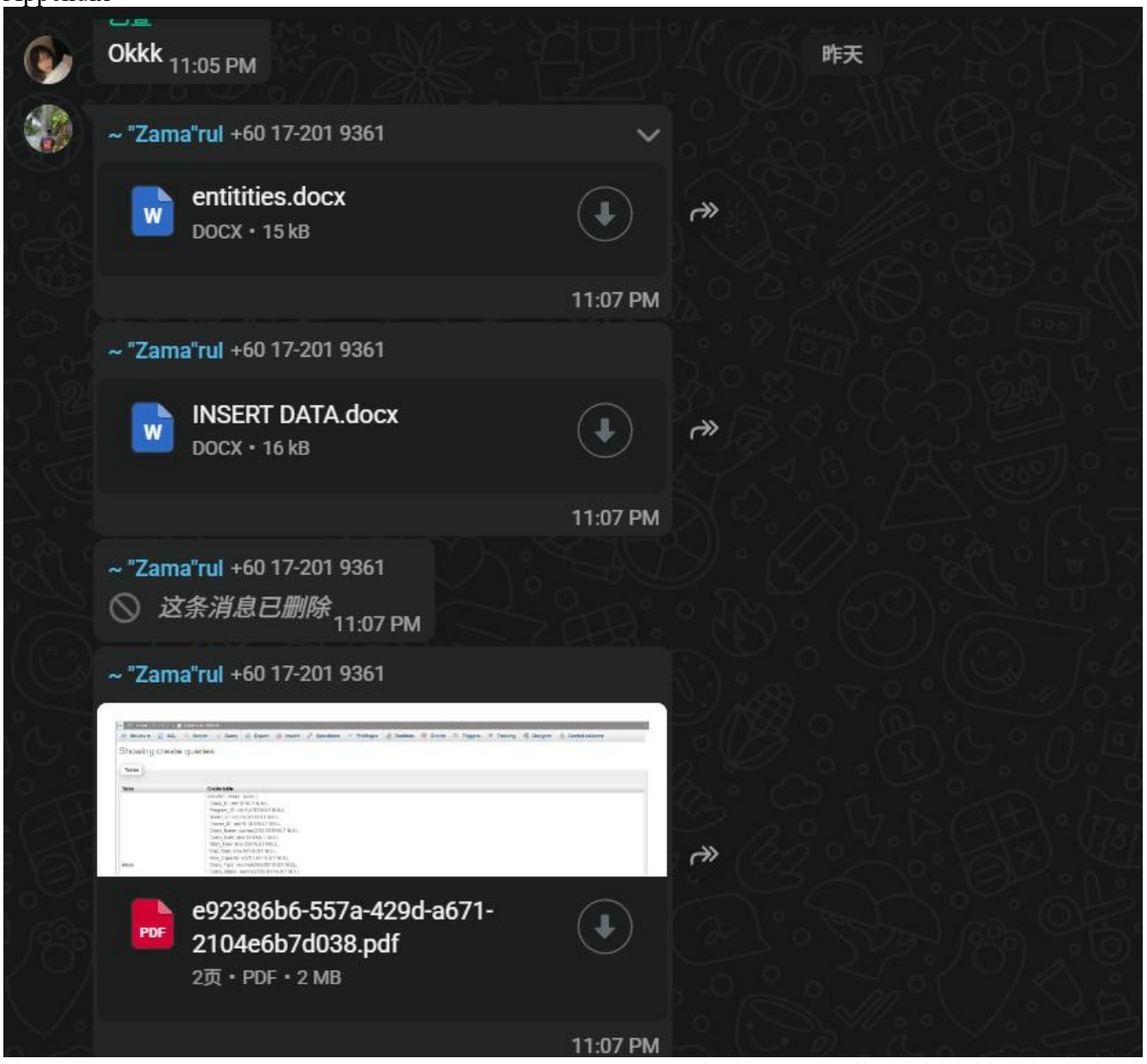
Home Back

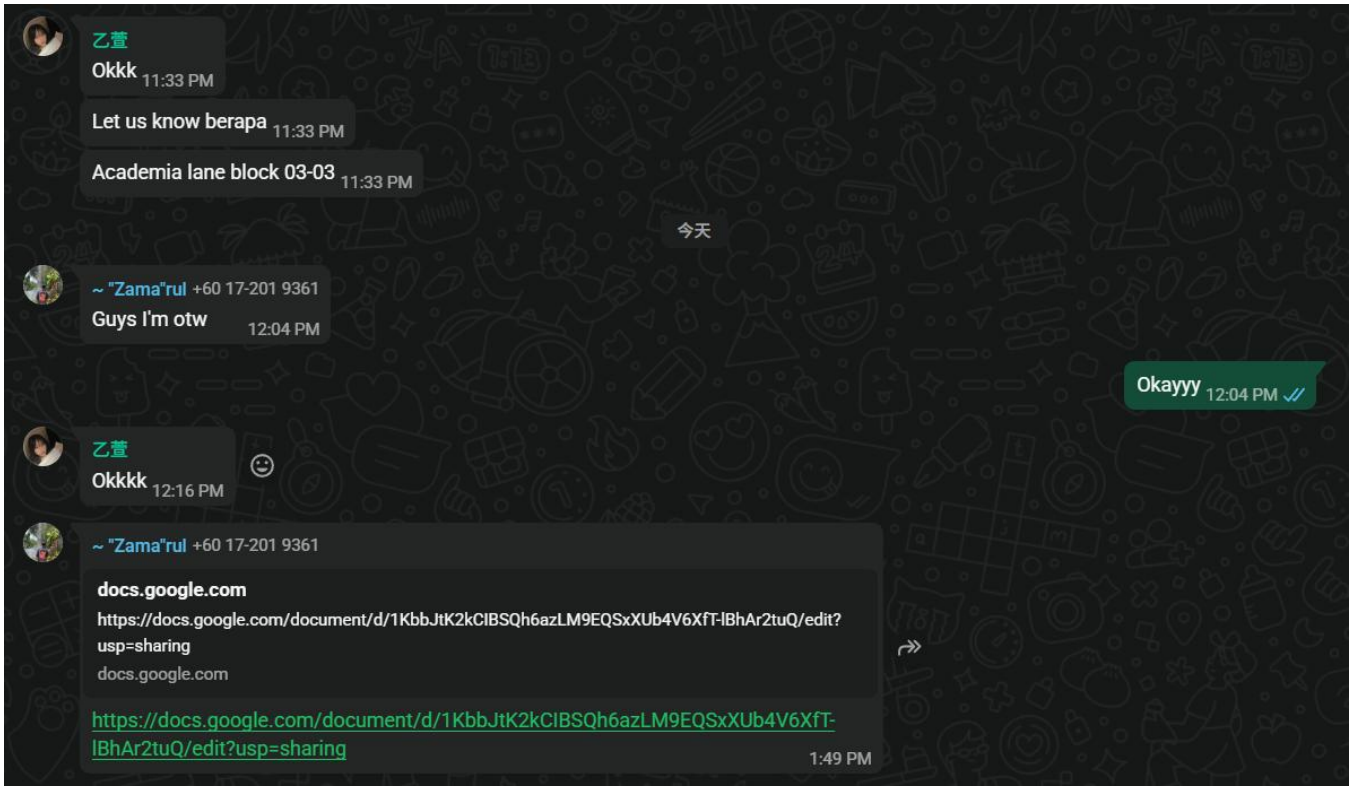
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User Manual

Step 1: Open Xampp
Step 2: Unzip file
Step 3: Fill in (-u root -p -v)
Step 4: Create database fitlifedb
Step 5: Use fitlifedb
Step 6: Copy the whole (sql script for create table and dummy data)
Step 7: Paste the copied script xampp
Step 8: Copy the fitlife db zip files that have been extraced to htdocs under xampp folder
Step 9: Open a website and type in (http://localhost/fitlifedb/)
Step 10: Run the program yay 😊

Appendix




乙萱
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Let us know berapa 11:33 PM
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Relational Database Diagram (15 marks) – CLO2						
Criteria	**1**	**2**	**3**	**4**	**5**	**Mark**
Relational database diagram (5 marks)	Relational database diagram is mostly incomplete and incorrect	Relational database diagram is partly complete but contains major missing attributes	Relational database diagram is mostly complete but has some missing or improperly defined attributes	Relational database diagram is complete but has minor inappropriate key attributes	Relational database diagram is complete, with key attributes appropriately assigned	
Time-Variant Data (2 marks)	Ability to capture data whose values change over time is partially working	Ability to capture data whose values change over time is working				
Relationship & Cardinality (3 marks)	Relationship and cardinality are mostly assigned incorrectly	Relationship and cardinality are partially assigned correctly	Relationship and cardinality are correctly assigned			
Normalized database (5 marks)	Major issues with redundancy, partial dependencies, and transitive dependencies	Some tables (>50%) fail to achieve 3NF, with partial or transitive dependencies causing major redundancy	Tables (>70%) mostly follow 3NF, but some partial or transitive dependencies exist, leading to minor redundancy and affecting overall integrity	Most tables (>80%) follow 3NF with minor issues in redundancy, but they do not affect overall integrity	All tables strictly follow 3NF. No redundancy, and all non-key attributes are fully functionally dependent on the primary key (s)	
Implementation of database application (25 marks) – CLO4						
Criteria	**1**	**2**	**3**	**4**	**5**	
Functionality (9 marks) — **Add (3 marks)**	Able to insert data for some entities but with major errors	Able to delete data for most entities but with minor errors	Able to insert data for all entities effectively			
Functionality (9 marks) — **Delete (3 marks)**	Able to delete data for some entities but with major errors	Able to delete data for most entities but with minor errors	Able to delete data for all entities effectively			
Functionality (9 marks) — **Update (3 marks)**	Able to update data for some entities but with major errors	Able to update data for most entities but with minor errors	Able to update data for all entities effectively			
Data Manipulat — **Query (Accuracy) (5 marks)**	Manipulate most required data components poorly	Manipulates some required data components but with major inaccuracies	Manipulates most required data components, but some have minor inaccuracies	Manipulates all required data components, with minor inaccuracies	Manipulates all required data components accurately	

	Query (Optimizati on) (3 marks)	Queries are mostly not optimized	Queries are partly optimized	Queries are efficiently optimized			
	Many-to-many data (3 marks)	Able to capture many-to-many data (>25%)	Able to capture many-to-many data (>50%)	Able to capture many-to- many data (>90%)			
	Dummy data (2 marks)	Data quantity is sufficient for basic functionality and manipulation	Sufficient quantity of data provided to test all functionality and manipulation				
Creativity (5 marks)		The application integrates one creative feature that provides minimal enhancement to its usability	The application integrates one creative feature that moderately enhances its usability	The application integrates one creative feature that greatly enhance its usability			
Presentation (10 marks) – CLO4							
Criteria		**1**	**2**	**3**	**4**	**5**	
Demonstration of Application (5 marks)		Demonstration of required functionality and data manipulation is unclear	Demonstration of required functionality and data manipulation is mostly unclear	Demonstration of required functionality and data manipulation is partly clear	Demonstration of required functionality and data manipulation is clear	Demonstration of required functionality and data manipulation is very clear	
Q&A (5 marks)		Responds poorly to questions or provides irrelevant responses	Responses are unclear or provide incomplete answers	Responds adequately to questions but shows minor gaps in understanding	Responds to most questions with good understanding	Responds effectively to all questions, demonstrating thorough understanding of the application	
Total = 50 marks							